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UTILITY OPTIMISATION IN A BREWERY PROCESS BASED ON ENERGY INTEGRATION METHODOLOGY

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Abstract *This paper presents a methodology aimed at improving the energy efficiency of a brewery applying process integration techniques. The different steps of the analysis are presented. The first step is the identification of the process energy requirements and the corresponding heat loads, which allows the definition of the process hot and cold streams. The Pinch Analysis of the brewery reveals a heat recovery potential of 36% by improving the heat exchanger system. In order to satisfy the minimum energy requirements, optimal energy conversion technology configurations are calculated, taking into account economic and environmental criteria. The integration of suitable utilities is considered (cogeneration engine combined with heat pumping and refrigeration systems) and the interaction between them is analysed. In addition, a thermo-economic optimisation is performed in order to determine the optimal heat pump operating temperatures, specific to the case study. The results presented are valid for the French case and show the opportunity to reduce by 36% the brewery heating bill and by 44% the CO_2 emissions through the set up of an optimised utility configuration when compared to the current one. In addition, the optimal integration shows that the water consumption of the refrigeration can be suppressed and appropriately be replaced by a heat pumping effect. The results are confronted to the German case, where grid electricity generation and energy costs differ significantly, leading to contrasting conclusions. The process system analysis shows that when considering the recovery of the plant organic waste, bio-methane can be produced and valorized in the cogeneration engine. Eventually, it is demonstrated that the process can become self sufficient in terms of energy and drastically reduce its water consumption.*

Keywords: *process integration, pinch analysis, brewery, thermo-economic optimisation*

Nomenclature

COP	Coefficient Of Performance [-]
MER	Minimum Energy Requirement [kW]
$\dot{Q}_{th}$	Heat Load [kW]
M	Mass[kg]
$\widetilde{M}$	Molar Mass [kg/kmol]
v_{CH_4}	Methane Content of Brewery Waste [m^3/kg]
LHV	Lower Heating Value [MWh/kg]
$\widetilde{v}$	Molar Volume of Perfect Gases [m^3/kmol]
T	Temperature [K]

1 INTRODUCTION

Beer production (28Mt in 1999, EU-15) ranks 5th in European food industry and the European beer is widely exported. Breweries use significant amounts of water and energy to produce this fresh and tasty drink. In the current trend of high energy price, energy efficiency improvement of industrial processes represents an important way to reduce production costs. In addition, growing environmental concern encourages companies to consider innovative solutions not only to reduce the carbon footprint but also the water consumptions.

In the Top-Down approach developed by D. Muller et al. ([8], 2007) for analysing the energy efficiency of industrial processes in the food industry, Pinch Analysis is used to identify the possible heat recovery by heat exchange between the streams to be cooled down and the streams to be heated up. Pinch Analysis ([5]) targets the minimum heat requirement of a process through the graphical representation of the process energy requirements, called *composite curves*, and describes how it is possible to achieve the determined energy targets with a correctly designed network of heat exchangers. The systemic approach consists of the identification and characterisation of the main process unit operations (PUOs). The Top-Down approach ([8], 2007) shows that more than 80% of the energy consumption can be explained by describing only 20% of the units of a factory. For these important PUOs, models are used to characterise the set of hot and cold streams that are needed to achieve the operation. The choice of the minimum approach temperature ΔT_{min} allowed by the heat exchangers enables the determination of the process Minimum Energy Requirements in heating and cooling. Grand composite curve analysis helps towards identifying opportunities for energy-efficient utility integration to satisfy the energy requirements, such as combined heat and power (CHP) systems or heat

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pumps (see for example [3], 2009). Optimal utility integration can be achieved using a Mixed Integer Linear Programming (MILP) formulation, as described by Maréchal and Kalitventzeff ([6], 1998).
This paper presents the implementation of process integration methodology on a brewery. Section 2 presents the identification and the modelling of the main process unit operations and the Pinch Analysis of the brewery, allowing the determination of the process energy targets. The opportunity of integrating CHP systems and heat pumps is studied in section 3. A multi-objective optimisation is undertaken in order to determine a set of optimal heat pump operating temperatures, minimising the operating and investment costs, and taking into account environmental criteria. The influences of the unit cost and production mix of electricity are studied quantitatively through the comparison between the French and the German cases. It is also shown that brewery organic waste can be recoved for biomethanation and allows increased economic and environmental savings.

2 PINCH ANALYSIS OF A BREWERY

2.1 Process Description

The brewery studied corresponds to a typical brewing process. The target temperatures of the streams and the proportions of ingredients are determined by the product recipe. The brewing house is associated with beer production and is split into two parts:

- **a hot part**, described by the block flow diagram in Figure 1, called *mashing*, where the blend of water and malt (*Mash*) is firstly brewed at high temperature (76°C) so that the activated enzymes transform malt starch into sugar. The *Mash* is then filtrated to obtain wort, which is boiled with hops to develop beer flavour. Wort boiling is an energy intensive operation. The wort is clarified in a whirlpool to remove the hops and eventually cooled to the pitching temperature.

- **a cold part**, illustrated in Figure 2, mainly consisting of the wort fermentation by yeast, at constant temperature (11°C), during 2 weeks. The beer is then chilled (-2°C) and clarified before being stored in insulated tanks where it ends its maturation.

The rest of the process consists of the bottling of the beer produced in the brewing house. In the process studied, four conditioning lines are designed to bottle the beer in new bottles, in kegs and in returnable bottles that are washed beforehand. The bottles filled with beer are then pasteurised.
A Cleaning in Place (CIP) system with effluent recovery, designed to wash the tanks, is also modelled in the study.

Figure 1: *Block Flow Diagram of the Hot Part*

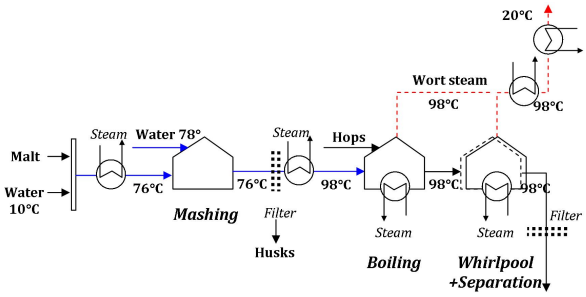
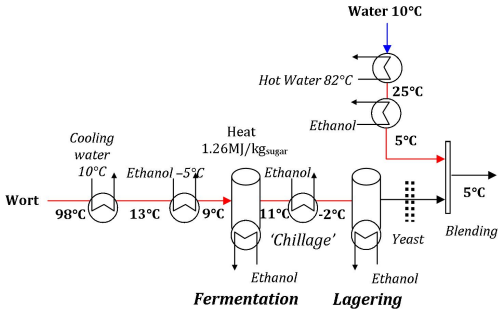


Figure 2: *Block Flow Diagram of the Cold Part*



2.2 Study Hypotheses

The Pinch Analysis of the brewery is performed using the following key hypotheses:

- Thermal losses during heat transfers are not taken into account.

- All the units are assumed to operate simultaneously. Thus, heat exchange is possible between different units. The yearly operating time of the brewery is 4992h.

- For each stream, the corresponding $\Delta T_{min}/2$ was chosen according to the existing equipments. The associated values of $\Delta T_{min}/2$ may not be optimal; however they are used in the study, as they correspond with the existing heat exchangers available in the factory.

Table 1: *Chosen Values of* $\Delta T_{min}/2$

Stream State	$\Delta T_{min}/2$ [Δ°C]
Liquid	2.5
Evaporating	0.8
Condensing	1.7

2.3 Modelling of the Conditioning Lines

The opportunity of recovering heat from the conditioning lines is worth studying, since bottle washing and pasteurisation devices represent important energy consumers in breweries. In the process under study, the

conditioning lines account for more than 32% of the current heating demand, which reveals the importance of modelling and integrating these units when undertaking the Pinch Analysis of breweries.

As an example, the modelling of the bottle pasteurisation device is presented in the next paragraphs. The device is considered as a sequence of soaking baths transferring their heat to the beer bottles passing through them. The bottles are successively heated and cooled; the baths thus require respectively heating and cooling supplies in order to keep a constant temperature level.

The chosen model consists of the representation of the energy requirements of the different baths at their corresponding temperatures. This representation enables the determination of internal heat recovery potential, as well as between the baths and other process streams. In the study, permanent regime is considered. The current bottle pasteurisation device consists of ten baths maintained at constant temperature levels. The input and output temperatures of the bottles are respectively 8°C/281K and 30°C/303K. The main soaking bath is kept at 62°C/335K.

The computed heat loads of the different baths correspond to the sum of the bottle heat loads and the heat losses to the surroundings (from conduction-convection and from radiation). The composite curves associated with the device can thus be obtained and represented in a $(T - \dot{Q})$ diagram. Figure 3 shows the composite curves corresponding to the device operating on the production line n°4. 40000 bottles of 0.33L/unit are currently pasteurised per hour by the machine.

In Figure 3, the hot streams are associated with the baths in heat excess and the cold ones with those requiring heat. The grand composite curve corresponds to the enthalpy (heat) difference between the hot and cold streams for each temperature interval. It can be seen in Figure 3 that it is possible to reduce the energy requirements of the device by setting up heat exchangers between the matched baths that can be identified using the composite curves. Indeed, the *self-sufficient zones* shown in Figure 4 reveal the opportunity of internal heat exchange between the heated and cooled baths of the device, such as N°2 and N°10, N°3 and N°9, N°4 and N°7, as well as N°4 and N°8.

The chosen value of $\Delta T_{min}/2$ (2.5°C) leads to a pinch corrected temperature T^*_{pinch} =15°C (288.2K), which corresponds to the minimal bath temperature ($T_{pinch,cold}$ = 12.5°C= $T^*_{pinch} - \Delta T_{min}/2$). As a result, the device does not need external cooling, since it is possible to transfer all the heat excess from the hot streams to the cold ones. The chosen value of ΔT_{min} is an optimal one, since it is associated with the minimal MER feasible for the current device (220kW in heating and 0kW in cooling).

In addition, it can be noted that the device heat recovery potential is determined not only by the ΔT_{min}, but also by the number of baths and by their temperature levels. Thus, when designing a bottle pasteurisation device, a thermo-economic optimisation can be performed based on Pinch Analysis and taking into account these additional criteria, in order to maximise the internal heat recovery.

Figure 3: *Bottle Pasteurisation Device (Line 4) Composite Curves*

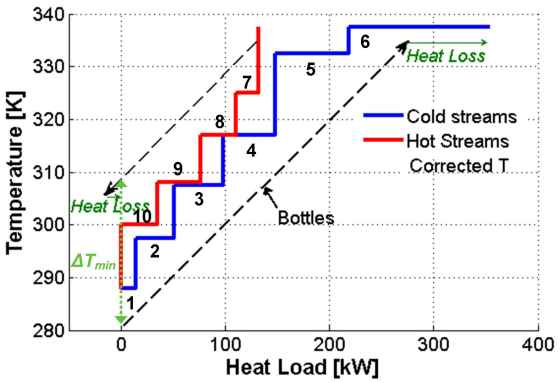
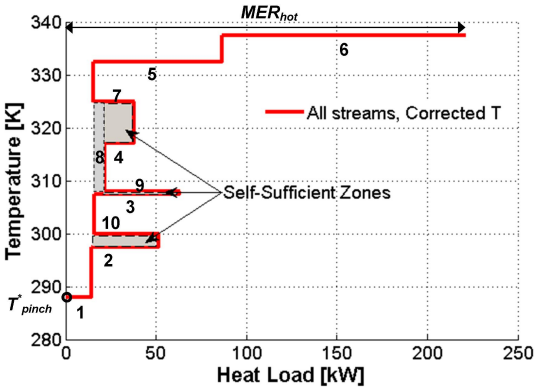


Figure 4: *Bottle Pasteurisation Device (Line 4) Grand Composite Curve*



2.4 Process Integration

Using the methodology and the study hypotheses described in section 1 and 2.2 respectively, the maximal heat recovery between the identified streams of the brewery can be quantified by computing the MER. Figure 5 presents the brewery composite curves resulting from the definition of the hot and cold streams identified in the process studied.

The first observation that can be established is related to the pinch, detected at the corrected temperature T^*_{pinch} =285.5K (or 12.5°C). The pinch point coincides with the temperature of cold water considered as the ambient temperature, above which the process needs external heating. As a consequence, all the effluents (hot streams) leaving the process at a temperature above the ambient temperature must deliver their heat to the cold streams beforehand.

The computation of the MER enables the identification of opportunities for energy saving. The resuls are shown in Table 2 and reveal a heat recovery potential of **1143kW**, which represents 36% of the total heating

consumptions, but will depend on the heat exchange efficiency characterised by the chosen value of ΔT_{min}.

Figure 5: *Brewery Process Composite Curves*

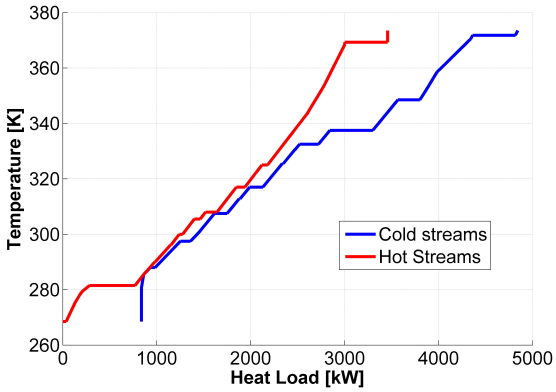


Figure 6: *Brewery Process Grand Composite Curve*

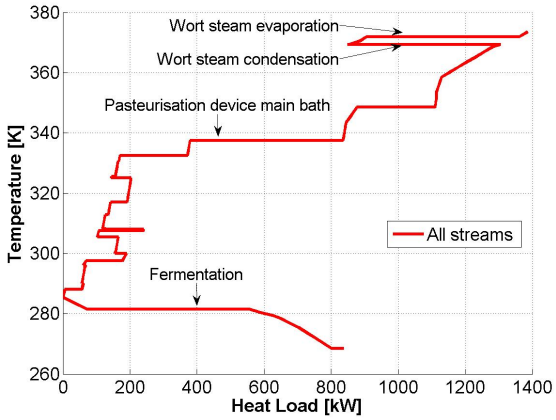


Table 2: *Estimated MER for* $\Delta T_{min,liquid}$ *=5°C*

Type	MER [kW]	Current Requirements [kW]	Savings [kW]	Savings
Hot Utility	**1386**	2529 $_{identified}$ 3133 $_{total}$	**1143**	45% **36%**
Cooling Water (>10°C)	**0**	NA	NA	NA
Refrigeration (<10°C)	**837**	NA	NA	NA

3 UTILITY INTEGRATION OPTIMISATION

This section presents the methododology developed in order to guide the choice of investors towards an optimised configuration of energy conversion systems designed to satisfy the MER determined in the previous section.
The energy costs taken into account in the study are expressed in euros (€) and are applied in the French industrial sector in 2007 [9]. The CO_2 emissions [2] are also considered as an optimisation criterion .

Table 3: *Energy Costs and CO_2 Emissions- France*

Energy /Resource	Unit Cost 2007 (Without Taxes)	CO_2 Emissions
Electricity	0.0541€/kWh$_e$	55g$_{CO2}$/kWh$_e$
Natural Gas	0.0271€/kWh$_{LHV}$	231g$_{CO2}$/kWh$_{LHV}$
Water	0.00657€/m^3	-

The French energetic context is compared to the German case, where energy management differ from France and is worth confronting.

Table 4: *Energy Costs and CO_2 Emissions- Germany [10]*

Energy /Resource	Unit Cost 2007 (Without Taxes)	CO_2 Emissions
Electricity	0.0927€/kWh$_e$	624g$_{CO2}$/kWh$_e$
Natural Gas	0.0417€/kWh$_{LHV}$	231g$_{CO2}$/kWh$_{LHV}$

3.1 Analysis of the Grand Composite Curve

The first step of the utility integration is the analysis of the process grand composite curve, which provides information on the temperature levels and the heat loads that must be met by the energy conversion technologies.
The analysis of the process grand composite curve leads to the following observations:

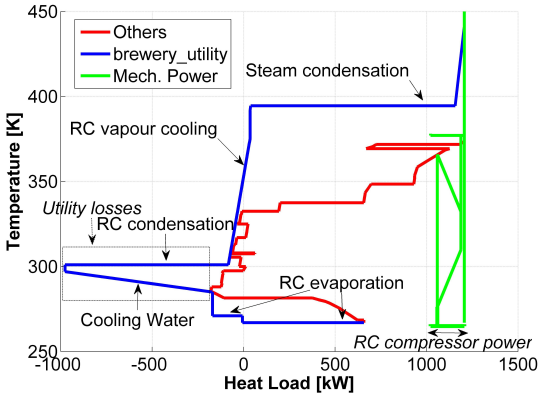
- The relatively low temperature levels of the process offer the opportunity to integrate combined heat and power (CHP) and heat pumping systems.
- The pinch temperature (T^*_{pinch} =12.5°C) corresponding to the ambient conditions is suitable for the integration of a refrigeration cycle across the pinch. Through the use of mechanical power, it is thus possible to pump the excess of heat from below the pinch to a higher level of temperature, where the process is lacking heat.
- An MVR system can fulfil part of the heating demand through the condensation of wort steam at high temperature, which would enable lower temperature heating requirements to be satisfied by the cooling water of a cogeneration engine. Thus, the size of the MVR system is determined in conjunction with the heat provided by a CHP system.
- A refrigeration utility with multiple levels of evaporation represents an appropriate solution minimising the exergy losses below the pinch temperature.

Using MILP formulation, the utility streams can be adjusted to the process streams in particular by the variation of their flow rates, which offers opportunity to minimise the operating costs when satisfying the process energy requirements.

3.2 Improving the Current Utility Setup

Currently, conventional utilities are used to supply energy to the process. A boiler fed with natural gas generates steam at high pressure (8.5bar) that is successively dissipated using valves to reach the desired pressure before condensing (2.2bar/123.3°C). For the cooling supply, the factory uses cold water and an NH_3-refrigeration cycle with two levels of evaporation, at -4°C and -8°C.
Using the observations formulated in section 3.1, an optimisation is performed in order to improve the current utility setup by minimising the operating costs. A MILP algorithm is used and enables the selection of the best utility configuration amongst the systems considered, as well as the computation of the mass flow rates of the utility streams minimising the yearly operating costs. In practical terms, the stream heat loads of the energy conversion technologies are optimised and added to the process hot and cold streams, resulting in the definition of the *integrated composite curves*.
It is important to note that the definition of the utility operating conditions (pressure, temperature) are to be set beforehand, since the sizes of the utilities computed by the MILP algorithm are determined as a function of them.

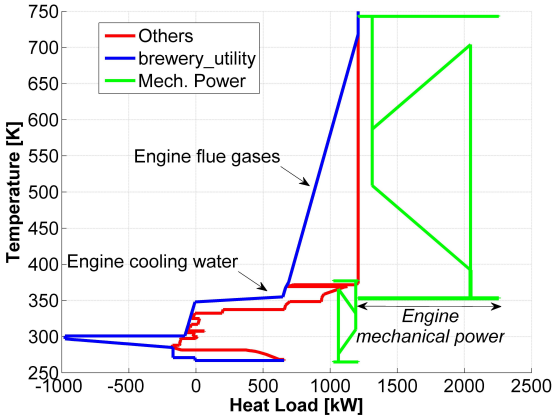
Figure 7: *Current Utility Setup: Boiler & Refrigeration Cycle (RC)*



In Figure 7 are presented the integrated utility composite curves associated with the current utility setup. The utility streams are represented by the line "brewery_utility" and the process requirements correspond to the grand composite curve "Others". The mechanical work supplied to compressors (heat pump and refrigeration cycle) is represented by the line "Mech. Power". The analysis of Figure 7 reveals that the current utility configuration does not prove optimal for multiple reasons. On the one hand, the use of steam at high pressure and temperature generates exergy losses, since the process requires heat at lower temperatures. On the other hand, it can be seen that below the condensation temperature of the refrigeration cycle, the process heating requirements are lower than the heat provided by the condensation of the refrigerant. This excess of heat must be evacuated by cooling water, which results in utility losses. It is therefore necessary to consider solutions allowing the improvement of the current utility configuration.

In order to reduce the exergetic losses due to the use of high pressure steam, the integration of a cogeneration engine is considered as an alternative to the boiler currently in operation. CHP systems represent a sustainable solution known to reduce the operating costs, as they produce both heat and mechanical power by taking advantage of fuel combustion. A reciprocating engine is considered in this study. It appears to be the most relevant technology, as it is possible to recover heat from both exhaust gases and cooling water, which can be used in low temperature processes like breweries. Natural gas, one of the most common fuels, is firstly considered. As can be seen in Figure 8, the exaust gases enable wort evaporation (T*=373K), whereas the engine water provides heat to the process streams below 360K. Fuel transformation leads to the generation of 1047kW of mechanical power. Part of this power can be used to run the refrigeration cycle compressors, which represents an important reduction in process electricity bill.

Figure 8: *Boiler Replaced with a CHP System*



However, the size of the CHP system can still be optimised and the utility losses caused by refrigeration cycle condensation remain a problem that has not been solved yet. To remedy these issues, the integration of heat pumps is eventually considered. As seen previously (see section 3.1), an MVR system can take advantage of the condensation heat provided by the steam emanating from the wort boiler and enables CHP system size optimisation. In addition, a high temperature heat pump permits utility loss reduction by making the condensation heat of the refrigeration cycle available at higher temperature levels.
The choice of heat pump operating conditions may have a high impact on the operating costs and on the sizes of the other utilities. In order to determine optimal heat pump operating temperatures, a multi-objective, thermo-economic optimisation is performed,

integrating the purchase cost of new utilities. Three decision variables are considered: the heat pump condensation temperature, the refrigeration cycle condensation and high pressure evaporation temperatures. Using the evolutionary algorithm *QMOO* developped by G.B. Leyland ([4], 2002) and A. Molyneaux ([7], 2002) a family of Pareto-optimal points is obtained, representing the trade-off between increased investment costs and lower operating costs. For the brewery studied, the population of Pareto-optimal points (see Figure 10) is divided into two distinct clusters, characterised by a single value of the heat pump condensation temperature (see Figure 9), namely 66.5°C(340K) for cluster 1 and 77.5°C(351K) for cluster 2, which corresponds to the maximal temperatures of the conditioning line units.

Figure 9: *Pareto Front (84 Pareto-optimal points after 2000 iterations)*

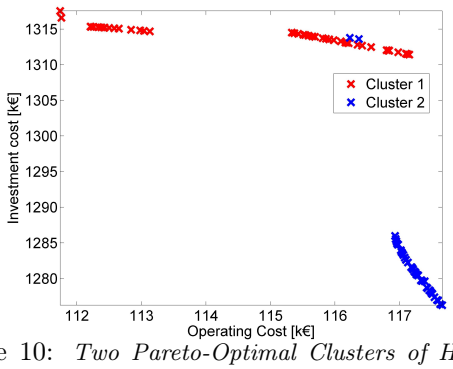
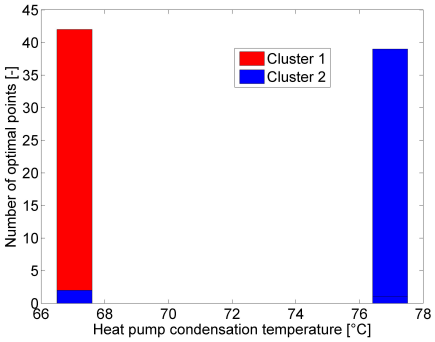


Figure 10: *Two Pareto-Optimal Clusters of Heat Pump Condensation Temperatures*



As for the refrigeration cycle, the evolutionary algorithm has converged towards an optimal value of 6°C (299K) for the high pressure evaporation temperature. The condensation temperatures are preferably distributed between 45°C (318K) and 50°C (323K).

Two optimised configurations, including the integration of MVR and heat pump systems, are presented in Figures 11 and 12. It can be seen a clear reduction of exergy losses: utility temperatures are as close as possible to the temperatures of the process energy requirements. One can also observe a drastic reduction in utility losses: for the case where the heat pump condenses at 77.5 °C (351K), external cooling water requirement is close to zero.

Table 5 presents the results associated with the different utility setup optimisation steps. It is considered that the remaining heating needs (604kW$_{th}$) of the non identified process units (NIPU) can be fulfilled by the use of the current boiler (considered efficiency=0.85%), fed by natural gas. The French case is confronted to the German case, where energy management is sensibly different (higher electricity price when compared to natural gas cost, electricity mainly produced by coal power plants, see Table 4). One can observe than in France, from both an economic and an environmental point of view, the most interesting utility setup consists of using preferably heat pumping systems with respect to CHP systems: the last configuration enables 36% of the operating costs and 18% of the total costs to be saved when compared to the current utility setup and permits a drastic reduction in CO_2 emissions (44%) and in water consumption. On the contrary, in Germany, high electricity price when compared to gas cost tends to promote cogeneration systems, which enable important reductions in operating costs and in CO_2 emissions. It is important to note that only energy costs are taken into account in the computation of the yearly operating costs. If carbon taxation was considered, the most environment-friendly setups would be associated with increased economic savings.

Figure 11: *CHP System+MVR, Heat Pump Condensing at 66.5°C, COP=5.37*

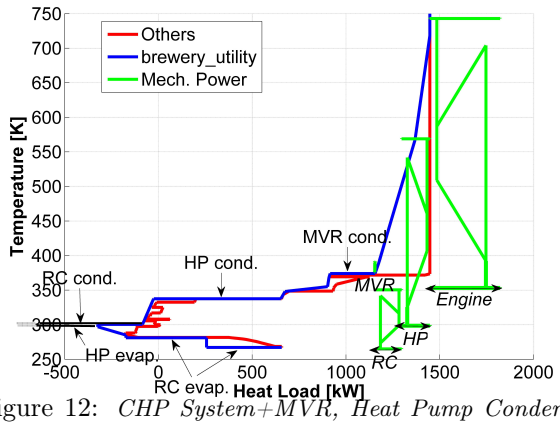


Figure 12: *CHP System+MVR, Heat Pump Condensing at 77°C, COP=5.71*

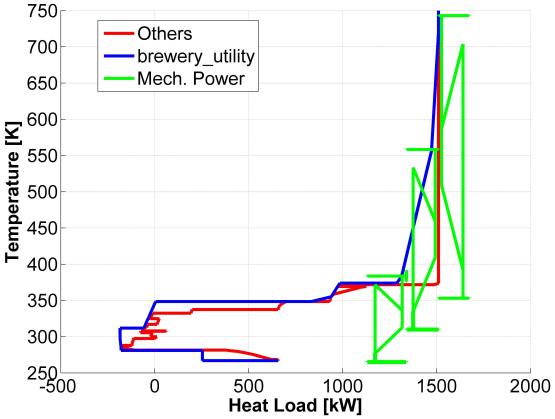


Table 5: *Results*

	Unit	1.MER & boiler	2.MER & CHP	3.MER & CHP & MVR & HP T_{cond}=66.5°C	4.MER & CHP & MVR & HP T_{cond}=77.5°C
Fuel consumption	[kg/s]	0.044	0.068	0.035	0.024
Electricity	[kWe]	184	-863	-80	142
Operating Costs FR	[k€/year]	332	210	205	212
Saving potential	*[%]*	*0*	*-37*	*-38*	*-36*
Operating costs GER	[k€/year]	520	283	312	336
Saving potential	*[%]*	*0*	*-46*	*-40*	*-35*
TOTAL COSTS FR[4]	[k€/year]	**332**	**308**	**274**	**274**
Saving potential	*[%]*	*0*	*-7*	*-17*	*-18*
TOTAL COSTS GER	[k€/year]	**520**	**380**	**381**	**398**
Saving potential	*[%]*	*0*	*-27*	*-27*	*-24*
CO_2 (EDF mix)	[tons/year]	**2459**	**3544**	**1912**	**1372**
Saving potential	*[%]*	*0*	*+44*	*-22*	*-44*
CO_2 (GER mix)	[tons/year]	2987	1094	1686	1976
Saving potential	*[%]*	*0*	*-63*	*-44*	*-34*
Cooling water	[kg/s]	17.1	17.1	3.0	0.1
Saving potential	*[%]*	*0*	*0*	*-82*	*-99*

3.3 Husk Bio-Methanation

Breweries offer the opportunity of recovering energy through husk bio-methanation. The recovered biogas can be used as an alternative to natural gas to feed the cogeneration engine. Knowing the amount of husk produced per year, it is possible to calculate the primary energy that can be recovered :

$$Q_{LHV} = \frac{M_{husk} \times \widetilde{M}_{CH_4} \times v_{CH_4} \times LHV_{CH_4}}{\widetilde{v}}$$

75 m^3 of methane can be recovered from 1 ton of husk [1], which represents, for the brewery studied, 8287MWh$_{LHV}$/year=**1660kW**$_{LHV}$, corresponding to a CHP system of 677kW$_e$. The organic matter is blended and its transformation into biogas by microorganisms requires a specific operating temperature (35°C) [1], which results in additional electricity and heat consumptions.

Table 6: *Results Bio-Methanation*

	Unit	1.MER & boiler	2.MER & CHP	3.MER & CHP & MVR & HP T_{cond}=66.5°C	4.MER & CHP & MVR & HP T_{cond}=77.5°C
Biogas Engine Size	[kW$_e$]	0	-1232	-677	-677
Process Identified Elec.	[kW$_e$]	184	184	295	379
Digester Elec.	[kW$_e$]	80	123	80	80
Total Elec.	[kW$_e$]	264	-925	-298	-219
Biogas Extra Heat	[kW$_{th}$]	39	0	196	434
Boiler (NIPU)	[kW$_{LHV}$]	664	711	480	200
Operating Costs FR	[k€/year]	161	-31	-16	-32
Operating Costs GER	[k€/year]	260	-280	-38	-60
Invest. Biomethanation	[k€]	895	2030	1418	1418
Invest. Heat Pumps	[k€]	0	0	198	290
TOTAL COSTS FR	[k€/year]	**238**	**145**	**124**	**115**
Savings /ref.	*[%]*	*-28*	*-56*	*-63*	*-65*
TOTAL COSTS GER	[k€/year]	**338**	**-105**	**101**	**88**
Savings /ref.	*[%]*	*-35*	*-120*	*-81*	*-83*
CO_2 (EDF mix)	[tons/year]	**839**	**566**	**471**	**170**
Savings /ref.	*[%]*	*-66*	*-77*	*-81*	*-93*
CO_2 (GER mix)	[tons/year]	**1588**	**-2060**	**-377**	**-452**
Savings /ref.	*[%]*	*-47*	*-169*	*-113*	*-115*

Table 6 presents the comparison between the different options determined throughout the utility integration optimisation procedure. Again, the calculated energy consumptions include the NIPU heat loads that are satisfied by the current boiler supplied with biogas. The reference investment cost of a biogas installation

[4]Total Yearly Costs = Operating Costs+Annualised Investment (interest rate=5%, payback time=15 years)

is 450k€(digester+cogeneration of 100kW$_e$) [1].
Table 6 reveals that biomethanation on site is the most economic and environmental solution for the brewery under study, when compared to the use of conventional energy sources. Indeed, the investment in a biomethanation installation is highly profitable and make the process become self-sufficient in terms of fuel. In France, bio-methanation allows the yearly total energy bill to be lessen by 65% and 2'289tons of CO_2 to be saved (93% with respect to reference) when choosing the most profitable setup in the French case, i.e. n°4. Again, the results are very dissimilar between France and Germany. The high cost and CO_2 content of German grid electricity promotes all the more the use of cogeneration run by biogas, which results in important economic and environmental profits.

4 CONCLUSION

A methodology based on process integration techniques has been presented, aimed at improving the energy efficiency of a brewery.
The definition and the modelling of the identified process units allows the determination of the heat recovery potential between process streams, using Pinch Analysis and linear programming. Then, the analysis of the process composite curves enables a first identification of utilities that can be used to fulfill the determined Minimum Energy Requirements.
The methodology presented combines utility integration with multi-objective optimisation in order to select the best utility setup and the corresponding operating conditions so as to minimise the operating and investment costs. It has been shown that integrating combined heat and power system together with heat pumps can be profitable from both an economic and an environmental point of view. A special focus is made on the dependance on the electricity cost and production mix. The confrontation between France and Germany is presented as an example: the contrasted electricity economic and environmental costs with respect to gas result in a solution promoting heat pumps in France whereas in Germany cogeneration systems prove more profitable. The quantification of costs and CO_2 emissions differences between the two cases is proposed.
The specificity of the process studied lies in the opportunity of recovering energy from brewery organic waste through bio-methanation. A quantitative analysis shows that the production and use of biogas on site lead to a drastic reduction in the total costs for both cases. However, the reduction in CHP system operating cost is not sufficient to substitute heat pumps by cogeneration if the brewery studied was located in France.
The energy requirements of the brewery are evaluated considering a continuous process functioning, which limits the accuracy of the results presented. Indeed, the identified units may not operate simultaneously, hence the interest of performing a multi-period analysis, which would require additional information on instantaneous mass fluxes. The quantitative results presented are specific to the brewery studied and it is important to keep in mind that any process has singularities that can hardly be transposed into another case study, without prior verification. However, the methodology implemented provides leads of solutions when seeking the improvement of brewery energy efficiency.

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